SHEAVES OF MODULES ON A RINGED SPACE HAVE FUNCTORIAL INJECTIVE EMBEDDINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the construction is displayed and each of its properties is either shown or attributed to a named result carried on this card as a navigation stub.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Functorial injective embeddings of modules. More on Algebra, Section more-algebra-section-injectives-modules. Supplies, for every ring and module over it, a functorial injective embedding of the module into an injective module.
- Pointwise products of sheaves. Sheaves, Examples sheaves-example-sheaf-product-pointwise and sheaves-example-sheaf-product-pointwise-algebraic-structure. Supply that the assignment sending an open set to the product of the chosen injective modules over the stalks is a sheaf of modules.
- Injectivity detected on stalks. Sheaves, Lemma sheaves-lemma-sheaf-subset-stalks and the description of injective sheaves of modules. Supply that a map of sheaves of modules is injective when it is so on stalks, and that a sheaf constructed stalkwise from injective modules is injective.
- Ringed spaces. Sheaves, Section sheaves-section-ringed-spaces. Supplies the structure sheaf and the stalks of the ringed space that the construction uses.

Source reference. Original section 0090: sources/stacks-sheaves/pinned-original-source.tex, source lines 3248–3248. Navigation pointer invoked at proof line 717; the counted span contains the pointer only and the cited result is not reproduced

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inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 01D8: sources/stacks-more-algebra/pinned-original-source.tex, source lines 13617–13617. Navigation pointer invoked at proof line 665; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 006X: sources/stacks-sheaves/pinned-original-source.tex, source lines 814–814. Navigation pointer invoked at proof line 667; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 007R: sources/stacks-sheaves/pinned-original-source.tex, source lines 1522–1522. Navigation pointer invoked at proof line 729; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0079: sources/stacks-sheaves/pinned-original-source.tex, source lines 1163–1163. Navigation pointer invoked at proof line 677; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 0099: sources/stacks-sheaves/pinned-original-source.tex, source lines 3487–3487. Navigation pointer invoked at proof line 689; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 009C: sources/stacks-sheaves/pinned-original-source.tex, source lines 3549–3549. Navigation pointer invoked at proof line 696; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 015Z: sources/stacks-homology/pinned-original-source.tex, source lines 7487–7487. Navigation pointer invoked at proof line 762; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0137: sources/stacks-homology/pinned-original-source.tex, source lines 7335–7335. Navigation pointer invoked at proof line 699; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space, see Sheaves, Section 0090. The category of sheaves of $\mathcal{O}_X$-modules on X has enough injectives. In fact it has functorial injective embeddings.*

Proof. For any ring R and any R-module M we denote $j : M \to J_R(M)$ the functorial injective embedding constructed in More on Algebra, Section 01D8. Let

$\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules on X. By Sheaves, Examples 006X and 007R the assignment

$$\mathcal{I} : U \mapsto \mathcal{I}(U) = \prod\nolimits_{x \in U} J_{\mathcal{O}_{X,x}}(\mathcal{F}_x)$$

is an abelian sheaf. There is a canonical map $\mathcal{F} \to \mathcal{I}$ given by mapping $s \in \mathcal{F}(U)$ to $\prod_{x \in U} j(s_x)$ where $s_x \in \mathcal{F}_x$ denotes the germ of s at x. This map is injective, see Sheaves, Lemma 0079 for example.

It remains to prove the following: Given a rule $x \mapsto I_x$ which assigns to each point $x \in X$ an injective $\mathcal{O}_{X,x}$-module the sheaf $\mathcal{I} : U \mapsto \prod_{x \in U} I_x$ is injective. Note that

$$\mathcal{I} = \prod\nolimits_{x \in X} i_{x,*} I_x$$

is the product of the skyscraper sheaves $i_{x,*}I_x$ (see Sheaves, Section 0099 for notation.) We have

$$\mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, I_x) = \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, i_{x,*} I_x).$$

see Sheaves, Lemma 009C. Hence it is clear that each $i_{x,*}I_x$ is an injective $\mathcal{O}_X$-module (see Homology, Lemma 015Z or argue directly). Hence the injectivity of $\mathcal{I}$ follows from Homology, Lemma 0137. □